# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 8 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 1

**Full Syllabus (Chapters 1–13)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** A student tests the effect of the mass of a bob on the time period of a pendulum. Which is the dependent variable? 1

**(a)** time period **(b)** place of testing
**(c)** mass of the bob **(d)** length of the string

**2.** A microscope has a 10x eyepiece and a 40x objective. A cell appears 2 mm wide in the image. The actual width is: 1

**(a)** 0.5 mm **(b)** 0.05 mm **(c)** 800 mm **(d)** 0.005 mm

**3.** A person has a fever, a rash and a headache. Which of these is a sign, not a symptom? 1

**(a)** feeling weak **(b)** headache
**(c)** body ache **(d)** rash seen by the doctor

**4.** A student connects a bulb, a switch and a cell but the bulb does not glow even when the switch is closed. The bulb holder is loose. The fault is a: 1

**(a)** magnetic effect **(b)** break in the circuit
**(c)** overload **(d)** short circuit

**5.** A block is pulled with 4 N but does not move. The friction acting on it is: 1

**(a)** 2 N **(b)** 4 N **(c)** 8 N **(d)** 0 N

**6.** A force of 240 N acts on an area of 0.4 square metre. The pressure is: 1

**(a)** 96 Pa **(b)** 600 Pa **(c)** 240 Pa **(d)** 60 Pa

**7.** A crystal of potassium permanganate is dropped into water in two beakers, one hot and one cold. The colour spreads faster in the hot beaker because: 1

**(a)** crystal dissolves into gas
**(b)** water gets heavier
**(c)** particles move faster at higher temperature
**(d)** cold water is thicker

**8.** Which of the following pairs contains only elements? 1

**(a)** $\mathrm{NaCl}$ and $\mathrm{Fe}$ **(b)** $\mathrm{O_2}$ and $\mathrm{Fe}$
**(c)** $\mathrm{CO_2}$ and $\mathrm{N_2}$ **(d)** $\mathrm{H_2O}$ and $\mathrm{O_2}$

**9.** 40 g of salt is added to 160 g of water. The mass percentage of the solution is: 1

**(a)** 20 per cent **(b)** 25 per cent **(c)** 10 per cent **(d)** 40 per cent

**10.** A ray strikes a plane mirror making an angle of 25 degrees with the mirror. The angle of reflection is: 1

**(a)** 50 degrees **(b)** 65 degrees **(c)** 25 degrees **(d)** 155 degrees

**11.** A vertical stick of 1 m casts a shadow of 1 m long at some time. The Sun's elevation at that time is about: 1

**(a)** 90 degrees **(b)** 45 degrees **(c)** 60 degrees **(d)** 30 degrees

**12.** In the food chain: phytoplankton, zooplankton, small fish, big fish, the secondary consumer is: 1

**(a)** zooplankton **(b)** big fish **(c)** phytoplankton **(d)** small fish

**13.** If the radius of the Earth is about 6400 km, the thickness of the crust (about 35 km) is: 1

**(a)** about 50 per cent of the radius **(b)** equal to the radius
**(c)** less than 1 per cent of the radius **(d)** about 25 per cent of the radius

**14.** After heating, the temperature readings are 40, 60, 80 and 100 degrees C at times 1, 2, 3 and 4 min. The rate of rise is: 1

**(a)** 20 degrees C per minute **(b)** 40 degrees C per minute
**(c)** 100 degrees C per minute **(d)** 10 degrees C per minute

**15.** A tissue differs from an organ because a tissue: 1

**(a)** has many types of tissues **(b)** forms systems
**(c)** has similar cells doing one job **(d)** is an organism

**16.** The correct sequence of events in infection by a pathogen is: 1

**(a)** symptoms, multiplication, entry **(b)** multiplication, symptoms, entry
**(c)** symptoms, entry, multiplication **(d)** entry, multiplication, symptoms

**17.** A toaster, an iron and a heater use the same effect of current, which is: 1

**(a)** chemical **(b)** heating **(c)** magnetic **(d)** light

**18.** A glass rod is rubbed with silk and brought near a suspended charged plastic rod, which is repelled. The plastic rod has: 1

**(a)** negative charge **(b)** no charge **(c)** magnetism **(d)** positive charge

**19.** Air pressure inside a bicycle tyre is greater than outside because: 1

**(a)** rubber is thick **(b)** tyre is hot
**(c)** the valve is open **(d)** more air is pumped in

**20.** A steam burn is worse than a burn from boiling water at the same temperature mainly because steam: 1

**(a)** is at a higher temperature than boiling water
**(b)** is heavier than water
**(c)** has a larger mass
**(d)** gives out extra heat when it condenses on the skin

**21.** Which of the following is NOT a metal? 1

**(a)** copper **(b)** sulphur **(c)** mercury **(d)** iron

**22.** A beam of light passes through a mixture and becomes visible. The mixture is a: 1

**(a)** gas **(b)** colloid **(c)** pure compound **(d)** true solution

**23.** A concave mirror of focal length 10 cm has a radius of curvature: 1

**(a)** 10 cm **(b)** 40 cm **(c)** 5 cm **(d)** 20 cm

**24.** A lunar year consists of 12 lunar months of 29.5 days each. The number of days is: 1

**(a)** 360 **(b)** 354 **(c)** 348 **(d)** 365

**25.** Which gas do aquatic plants release in the daytime that fish need? 1

**(a)** oxygen **(b)** methane **(c)** carbon dioxide **(d)** nitrogen

**26.** If air contains 21 per cent oxygen, in 500 L of air the volume of oxygen is about: 1

**(a)** 210 L **(b)** 105 L **(c)** 21 L **(d)** 78 L

**27.** A data table has more than one outlier. The best action is to: 1

**(a)** stop recording **(b)** check the experimental method
**(c)** use the biggest **(d)** drop all of them

**28.** A cell is stained in blue and the nucleus appears dark. The role of stain is to: 1

**(a)** make food **(b)** kill the cell **(c)** provide contrast **(d)** increase size

**29.** The immune system includes: 1

**(a)** bones and nails **(b)** teeth and hair
**(c)** white blood cells and antibodies **(d)** only skin colour

**30.** The magnetic effect of current is shown by a: 1

**(a)** bulb glow **(b)** bubble formation
**(c)** wire melting **(d)** compass needle deflection

**31.** Which of the following is NOT an effect of force? 1

**(a)** change in colour **(b)** change in speed
**(c)** change in direction **(d)** change in shape

**32.** A hurricane in the Atlantic and a typhoon in the Pacific are: 1

**(a)** different types of storms
**(b)** the same type of storm with different names
**(c)** earthquakes
**(d)** tornadoes

**33.** A gas at high pressure and low temperature is more likely to become: 1

**(a)** liquid **(b)** solid always **(c)** plasma **(d)** vapour only

**34.** A bell is made of bronze because it is: 1

**(a)** sonorous and hard **(b)** liquid
**(c)** brittle **(d)** soft

**35.** A student prepares tea with 5 g sugar in 100 mL and another with 10 g sugar in 100 mL. The second tea is: 1

**(a)** a suspension **(b)** saturated
**(c)** more dilute **(d)** more concentrated

**36.** A convex lens converges a parallel beam to a point called the: 1

**(a)** pole **(b)** principal focus **(c)** optical centre **(d)** axis

**37.** The Moon's waxing crescent is seen in the: 1

**(a)** west after sunset **(b)** east at midnight
**(c)** north at dawn **(d)** south at noon

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**38.** In a pond, the amount of dissolved oxygen falls when: 1

**(a)** plants grow **(b)** organic waste rots in it
**(c)** it rains **(d)** it is cleaned

**39.** The percentage of Earth's water that is fresh water is about: 1

**(a)** 30 per cent **(b)** 97 per cent **(c)** 50 per cent **(d)** 3 per cent

**40.** In a pendulum test of the effect of the mass of the bob on the time period, which must be kept constant? 1

**(a)** number of masses **(b)** length of the string
**(c)** mass of the bob **(d)** time period

**41.** A cell that has a nucleus, a cell wall and chloroplasts is likely from: 1

**(a)** blood **(b)** a leaf **(c)** human cheek **(d)** bacterium

**42.** In a village, many people fell ill with diarrhoea after a flood. The most probable cause is: 1

**(a)** mosquito bite **(b)** dust
**(c)** contaminated drinking water **(d)** sunlight

**43.** Two wires of equal length and thickness, one nichrome and the other copper, carry the same current. Which gets hotter? 1

**(a)** copper **(b)** nichrome **(c)** both equally **(d)** neither

**44.** A cart is pushed with 20 N and friction is 12 N. The net force is: 1

**(a)** 20 N **(b)** 8 N **(c)** 32 N **(d)** 12 N

**45.** A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The maximum pressure it can exert when resting on a face is: 1

**(a)** 200 Pa **(b)** 2 Pa **(c)** 500 Pa **(d)** 0.8 Pa

**46.** When 10 g of salt is dissolved in 100 mL of water, the volume of the solution is hardly more than 100 mL. This is because: 1

**(a)** salt becomes gas
**(b)** salt disappears
**(c)** particles of salt fit in the gaps between water particles
**(d)** water evaporates entirely

**47.** Which of the following pairs contains only compounds? 1

**(a)** $\mathrm{Fe}$ and $\mathrm{CO_2}$ **(b)** $\mathrm{N_2}$ and $\mathrm{CH_4}$
**(c)** $\mathrm{NH_3}$ and $\mathrm{CH_4}$ **(d)** $\mathrm{O_3}$ and $\mathrm{CO_2}$

**48.** How many grams of sugar are in 250 g of a 12 per cent sugar solution? 1

**(a)** 30 g **(b)** 3 g **(c)** 25 g **(d)** 12 g

**49.** The angle between the incident and the reflected ray is 80 degrees. The angle of incidence is: 1

**(a)** 160 degrees **(b)** 10 degrees **(c)** 80 degrees **(d)** 40 degrees

**50.** At noon the shadow of a stick is directly towards the north (in India). The Sun then is: 1

**(a)** to the north **(b)** below the horizon
**(c)** to the south of the observer **(d)** overhead

**END OF QUESTION PAPER** **Total: 50 Marks**